

PAPER

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Efficient Nd³⁺ sensitized Yb³⁺ emission and infrared-to-visible energy conversion in gallium nano-garnets

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Nanocrystalline RE₃Ga₅O₁₂ (RE = Gd, Y and Lu) garnets co-doped with 1 mol% of Nd³⁺ and 10 mol% of Yb³⁺ ions have been synthesized via a sol–gel method and their structural and luminescence properties studied. Upon laser excitation at 805 nm, an intense near-infrared emission from Yb³⁺ ions at 1025 nm is observed, which is due to efficient Nd³⁺ → Yb³⁺ energy transfer with a probability of about 90%. Lifetimes of the ⁴G_{7/2} and the ⁴F_{3/2} emitting levels of Nd³⁺ ions and the ²F_{5/2} excited level of Yb³⁺ ions have been evaluated upon pulsed laser excitations. The non-exponential nature of the decay curves has been studied to understand the multipolar interaction between Nd³⁺ and/or Yb³⁺ ions using the Inokuti–Hirayama model. Moreover, under 970 nm laser excitation, green upconverted luminescence is observed in Nd³⁺–Yb³⁺ co-doped RE₃Ga₅O₁₂ nano-garnets, which can be related to both phonon-assisted excited-state absorption and energy transfer upconversion mechanisms. The results show that the as-synthesized nano-garnets could be useful for infrared detectors, solar energy converters and green phosphor applications.

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1. Introduction

Trivalent rare earth (RE³⁺) ion doped crystals with emissions at around 1 μm have been attracting great attention for their potential applications in continuous wave and pulsed lasers, scintillators, optical bistability, solar energy harvesting in c-Si solar cells, optical sensors, *etc.*^{1–4} To obtain 1 μm emission, the trivalent ytterbium (Yb³⁺) ion is the most favorable with a series of advantages over other RE³⁺ ions such as longer lifetime, the absence of cross-relaxation or excited-state absorption, low quantum defect that leads to reduction in thermal load, wide absorption and emission bands that are necessary to generate ultra-short laser-pulses through a mode-locking technique, *etc.*⁴ The rich and unique energy level structure of RE³⁺ ions permits effective spectral modification of the emission through energy transfer between neighbouring RE³⁺ ions for enhancing the efficiency of solar cells.⁵ Moreover, Yb³⁺ emission can still be enhanced by sensitizing with appropriate RE³⁺ ion with optimized doping concentration through energy transfer processes. Among different RE³⁺ ions, Nd³⁺ ion is good sensitizer with intense absorption bands in UV-visible-NIR regions with variety of low cost pumping options and characterized by

a fast non-radiative relaxation decays to the ⁴F_{3/2} emitting level, whose luminescence gives rise to different emission bands in the near-infrared range from 890 to 1400 nm, quite favorable for absorption of the Yb³⁺ ions.^{4,6}

In this context, the selection of suitable host matrix to achieve efficient luminescence at around 1 μm through energy transfer has become a subject of interest. Even though RE³⁺ doped fluoride crystals are efficient luminescent materials due to their low phonon energy, high quantum yield and long radiative lifetimes, they exhibit less phase stability compared to oxide crystals.^{7–10} For example, Eu³⁺-doped YF₃ nanocrystals transformed from orthorhombic to hexagonal phase when pressure is increased from ambient (0 GPa) to 12 GPa.¹⁰ But oxide crystals have been proved as thermally, chemically and mechanically stable materials.^{11,12} In particular, garnets are the best host matrices among oxide crystals due to their high solubility of RE³⁺ ions, wide transparency from UV to mid-IR and relatively low-energy phonons. Explicitly, gallium garnets are best hosts for RE³⁺ ions compared to aluminium garnets since replacement of Al³⁺ by Ga³⁺ ions in the RE₃Al₅O₁₂ (RE = Gd, Y and Lu) crystals would decrease its band gap and thus might reduce defect states through anti-sites by lowering the conduction band below to the defect states which results in efficient luminescence.^{13–16} Moreover, lattice dynamics and *ab initio* studies of structural, elastic, and vibrational properties of gallium nano-garnets under high pressure indicate that these nano-garnets are mechanically stable up to about 90 GPa and then transformed to amorphous state beyond this pressure.^{17,18} It was also reported that Ho³⁺ doped gallium nano-garnets

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emits more intense emission compared to $\text{Ho}^{3+}:\text{Y}_3\text{Al}_5\text{O}_{12}$ (YAG) nano-garnets.¹⁹ Hence, it is interesting to investigate the structural and luminescence properties of $\text{Nd}^{3+}\text{--Yb}^{3+}$ co-doped gallium nano-garnets.

Structural and luminescence properties of $\text{Nd}^{3+}\text{--Yb}^{3+}$ ions doped $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ (GGG), YAG, Sc_2O_3 , Y_2O_3 , YAlO_3 , $\text{Al}_4\text{B}_2\text{O}_9$, and $\text{YAl}_3(\text{BO}_3)_4$ crystals have been reported.^{20–27} Jia *et al.*, observed the Nd^{3+} sensitized Yb^{3+} emission at around 1025 nm due to energy transfer process from Nd^{3+} to Yb^{3+} in GGG single crystal grown by Czochralski technique.²⁰ Lupei *et al.*, studied the effect of Yb^{3+} ion concentration on Nd^{3+} emission and $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer process in YAG, Y_2O_3 , and Sc_2O_3 transparent ceramics synthesized by solid state ceramic technique.^{21–23} It is known that Czochralski technique is a quite costly, time consuming, and requires high temperature. In addition, the control of composition is not an easy task when growing single crystals. On the other side, solid state ceramic technique involves repeated grinding, sintering under vacuum condition, more time consuming, and high temperature. The synthesis technique that has to be adopted in the present investigation is simple, low cost, lower reaction temperature, easier composition control and environmentally benign and successfully prepared the GGG, YGG, and LuGG nano-garnets in a single phase cubic structure.

Maia *et al.* reported the structural and luminescence properties of $\text{Nd}^{3+}/\text{Yb}^{3+}$ doped $\text{Al}_4\text{B}_2\text{O}_9$ nanocrystals.²⁶ Rai *et al.*, studied the upconversion luminescence in Nd^{3+} doped $\text{Ga}_{10}\text{Ge}_{25}\text{S}_{65}$ glass.²⁸ In spite of all these studies, there are no reports on Yb^{3+} ion sensitised Nd^{3+} emission *i.e.*, infrared to visible energy conversion in gallium nano-garnets. Upconversion inorganic nanocrystals have technological applications such as fluorescent labels for sensitive detection of bio-molecules, photochemical reaction, bio-imaging, optical encoding, display technologies, *etc.*^{29–32}

Hence, the present investigation aims to study on Nd^{3+} sensitized Yb^{3+} emission, as well as Yb^{3+} sensitized Nd^{3+} upconversion luminescence in gallium nano-garnets. The effect of Yb^{3+} co-doping on the structural, optical and luminescence properties of $\text{Nd}^{3+}:\text{RE}_3\text{Ga}_5\text{O}_{12}$ nano-garnets (RE = Gd, Y and Lu) have been studied. Luminescent decay curves for $^4\text{G}_{7/2}$ and $^4\text{F}_{3/2}$ levels of Nd^{3+} ions and $^2\text{F}_{5/2}$ level of Yb^{3+} ions are analyzed by the Inokuti–Hirayama model to know the kind of multipolar interaction responsible for energy transfer between Nd^{3+} and/or Yb^{3+} ions. The $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer efficiency for intense emission at 1025 nm has been evaluated. Upon 970 nm laser excitation, intense green upconverted luminescence is observed to the naked eye and the mechanisms involved in the upconversion process has been analysed by time-resolved spectroscopy.

2. Experimental details

The $\text{RE}_{3-x-y}\text{Nd}_x\text{Yb}_y\text{Ga}_5\text{O}_{12}$ (where RE = Gd, Y, and Lu; $x = 0.01$ and $y = 0.1$; labelled as GGG1Nd10Yb, YGG1Nd10Yb, and LuGG1Nd10Yb, respectively) nano-garnets were synthesized *via* sol–gel method.¹⁹ X-ray diffraction (XRD) patterns were measured on a diffractometer (PANalytical X'Pert Pro) using

$\text{CuK}\alpha_1$ radiation. High resolution transmission electron micrographs were measured (JEOL microscope 200 kV) to know morphology of synthesized samples. The diffuse reflectance spectra were measured by using a spectrophotometer (Cary 5000). The luminescence spectra were measured by exciting with cw tunable Ti:sapphire laser (Spectra Physics 3900S) pumped by a 10 W cw Ar^+ laser (Spectra Physics 2060-10 Beamlock). The emission was focused onto a 0.32 m monochromator (Jobin Yvon Triax 320) coupled with thermoelectric photomultipliers (Hamamatsu R5108 and H10330B-75). The luminescence decay curves were obtained by exciting with a pulsed (10 ns) optical parametric oscillator (OPO) (EKSPLA/NT342/3/VE) using a digital storage oscilloscope (LeCroy WS424) coupled to the detection system. The temporal evolution of green luminescence was measured with the OPO by exciting at 970 nm. All the spectra were corrected for the spectral response of the equipment and were carried out at room temperature.

3. Results and discussion

3.1. Structural and morphological properties

The powder XRD patterns of GGG1Nd10Yb, YGG1Nd10Yb, and LuGG1Nd10Yb nano-garnets, shown in Fig. 1, reveal that all of the reflections in the profiles are well-indexed to a single phase cubic structure belonging to $Ia\bar{3}d$ space group (no. 230, $Z = 8$). The diffraction data were refined by Rietveld fitting using the FULLPROF program.^{33,34} Table 1 shows the unit cell parameters as well as the corresponding reliability factors (χ^2 , R_p , R_{wp} , and R_{exp}), which make clear the goodness of the fitting, for co-doped nano-garnets, along with those for Nd^{3+} singly doped nano-garnets. Only small differences are found in amplitudes of a few peaks (see Fig. 1). Different features can be outlined from these data: (a) the crystalline structures of the gallium nano-garnets are very similar for 1 mol% Nd^{3+} single-doped or

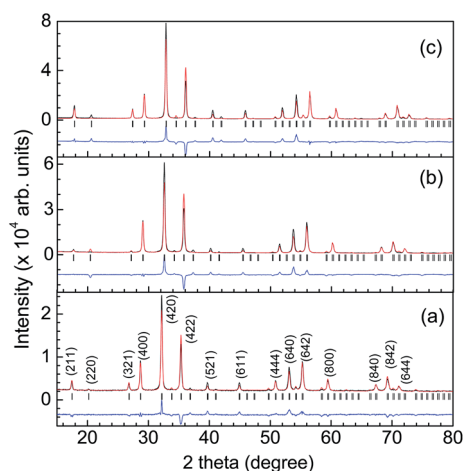


Fig. 1 X-ray diffraction pattern of (a) GGG1Nd10Yb, (b) YGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets. Rietveld refinements including the differences (blue) between calculated (red) and observed (black) patterns are also shown. The vertical lines are the allowed reflections for this material in the $Ia\bar{3}d$ (no. 230) space group.

Table 1 Unit cell parameters and reliability factors obtained from the XRD Rietveld refinement for 1Nd^{3+} and $1\text{Nd}^{3+}-10\text{Yb}^{3+}$:REGG nano-garnets

Compound	a (Å)	χ^2	R_p	R_{wp}	R_{exp}	Ref.
GGG1Nd10Yb	12.422	9.71	4.26	6.08	1.95	Present
YGG1Nd10Yb	12.299	6.92	12.4	16.6	2.00	Present
LuGG1Nd10Yb	12.192	8.73	14.0	18.5	1.98	Present
GGG1Nd	12.424	2.44	16.1	12.3	5.94	6
YGG1Nd	12.305	2.15	13.4	14.2	2.83	6
LuGG1Nd	12.189	2.01	16.9	18.2	2.71	6

$1\text{Nd}^{3+}-10\text{Yb}^{3+}$ co-doped and also quite close to those of their bulk counterparts (for example, $a = 12.19$ Å for bulk LuGG); (b) the decrease in unit cell parameter when the size of the RE^{3+} ion decreases ($\text{Gd}^{3+} > \text{Y}^{3+} > \text{Lu}^{3+}$) is due to the lanthanide's contraction; and (c) upon 10 mol% Yb^{3+} doping into the GGG1Nd, YGG1Nd and LuGG1Nd nano-garnets, one can observe that the cell parameters decrease with respect to the corresponding ones for GGG1Nd and YGG1Nd, respectively, due to the fact that the ionic radius of Yb^{3+} (0.99 Å) is smaller than the ionic radii of Gd^{3+} (1.05 Å) and Y^{3+} (1.02 Å). However, in the case of LuGG1Nd10Yb a slight increase of the cell parameters with respect to those for LuGG1Nd takes place due to the fact that the ionic radius of Yb^{3+} is slightly larger than Lu^{3+} (0.98 Å).

By using the Scherrer's formula,⁶ the average crystallite sizes of doped nano-garnets have been calculated and are found to be around 35 nm. In order to obtain nanoparticles in the form of pure garnet structure, the sol-gel synthesis needs thermal treatments that gives rise to particles of sizes not smaller than this value. When the size of the nanoparticles is very short (<5 nm) it is quite typical to observe distortions of the crystalline structure due to the high surface-to-volume ratio, which further affect the vibrational and optical properties of the material, but that disappear when the size increases.³⁵

The structural, vibrational and optical properties of nano-garnets are similar to their bulk counterparts,⁶ which may open new and interesting applications such as ultra-fast lasers, solar energy harvesters, infrared detectors and phosphors with increased resolution with respect to conventional bulk materials.¹⁻⁴ The differences in the ionic radii of Gd^{3+} (1.05 Å), Y^{3+} (1.02 Å) and Lu^{3+} (0.98 Å) ions have a dramatic effect on the cubic structure of the garnet by reducing the volume of the unit cell by 5%, an effect clearly observed in the shift of diffraction peaks to higher Bragg's angles when passing from Gd^{3+} to Lu^{3+} . Similar kind of effect was also observed in Nd^{3+} single-doped REGG nano-garnets.⁶

The morphology of these nano-garnets has been studied by high resolution transmission microscopy (HRTEM) and their micrograph is shown in Fig. 2 for YGG1Nd10Yb nano-garnets. As can be seen, the nano-garnets have been grown in the form of spheres with an average size of around 35–40 nm. The average size of the nano-garnets is well agreed with the crystallite size obtained from XRD technique. Similar results are observed for GGG1Nd10Yb and LuGG1Nd10Yb nano-garnets.

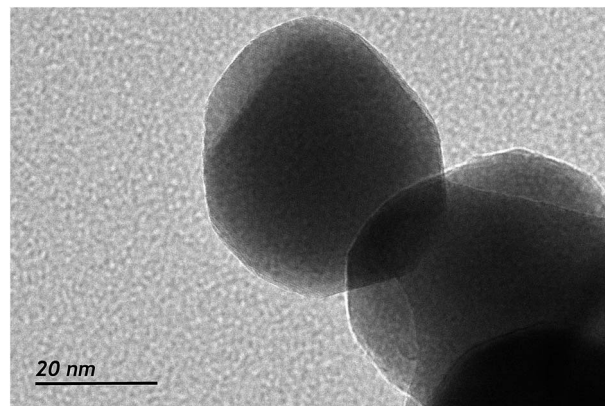


Fig. 2 HRTEM micrograph of YGG1Nd10Yb nano-garnets.

3.2. Diffuse reflectance spectra

Diffuse reflectance spectra were measured in the optical range from 400 to 1050 nm in order to identify the partial energy levels and local structure of optically active ions in the doped garnets [see Fig. 3(i) and (ii)]. The observed peaks between 400 and 900 nm are due to intra-configurational $4f^3-4f^3$ electronic transitions originating from the $^4\text{I}_{9/2}$ ground state to different excited levels of Nd^{3+} ion,⁶ whereas the peaks between 900 and 1050 nm are due to $4f^{13}-4f^{13}$ transitions of Yb^{3+} ions originating from different Stark levels of $^2\text{F}_{7/2}$ ground state to those of the single $^2\text{F}_{5/2}$ excited state of Yb^{3+} ion [see Fig. 3(ii)]. Transitions of Nd^{3+} and Yb^{3+} ions are assigned according to the well-known Dieke's diagram.³⁶ The sharp peak profiles are indicative of incorporation of Nd^{3+} and Yb^{3+} ions in the nano-garnet structure occupying the dodecahedral site of the RE^{3+} ($=\text{Gd}^{3+}$, Y^{3+} or Lu^{3+}) ions with a D_2 point symmetry. Fig. 4 shows the partial energy level diagram of Nd^{3+} and Yb^{3+} ions, together with radiative and non-radiative transitions.

When analyzing the crystal-field interaction it is worth noting that the D_2 local symmetry at the Nd^{3+} and Yb^{3+} sites in garnets is often described as an orthorhombic distortion of a main cubic symmetry. Thus, two contributions must be taken into account when describing the crystal-field interaction: a cubic symmetry and a superimposed orthorhombic symmetry component that accounts for the real D_2 symmetry at the Nd^{3+} and Yb^{3+} sites. Using this description, the crystal-field symmetry splits the ground $^2\text{F}_{7/2}$ multiplet into four levels and $^2\text{F}_{5/2}$ level into three levels (see Fig. 4).⁴ As can be seen from Fig. 3(i), the relative intensities and the fine-structure of the bands are similar for the three nano-garnets, but with slight differences in the energies of the fine structure of peaks due to changes in the bond distances and angles, and hence in the crystal-field strength, around Nd^{3+} and Yb^{3+} ions, similar to Nd^{3+} single-doped nano-garnets.⁶ These effects can be ascribed to chemical pressure around Nd^{3+} and Yb^{3+} ions, *i.e.* the smaller the ionic radius of the RE^{3+} ion the larger the crystal-field strength. In the case of the garnet structure, the increase of the crystal-field magnitude gives rise to a decrease of the orthorhombic crystal-field contribution.⁶ In addition, as shown in Fig. 3(ii),

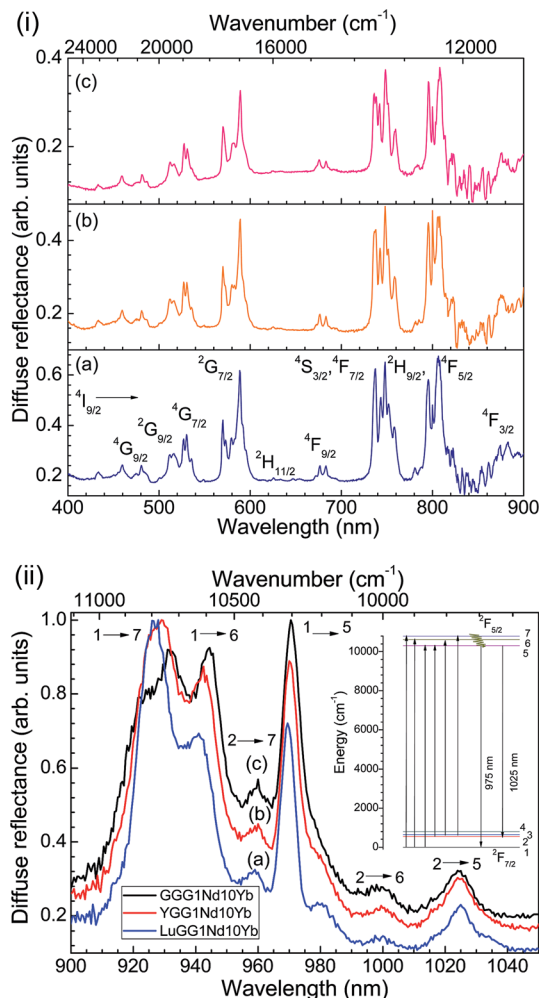


Fig. 3 (i) Nd^{3+} ion diffuse reflectance spectra of (a) GGG1Nd10Yb, (b) YGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets, (ii) Yb^{3+} ion diffuse reflectance spectra of (a) GGG1Nd10Yb, (b) YGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets. Inset shows Yb^{3+} absorption and emission channels in the present nano-garnets.

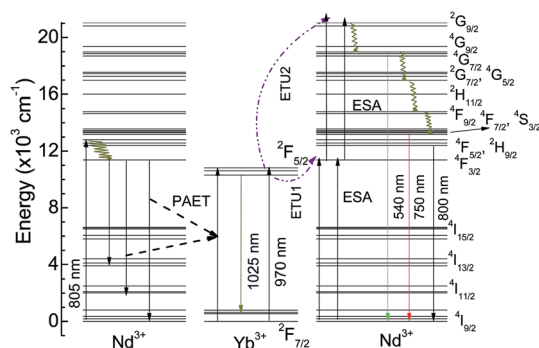


Fig. 4 Partial energy level diagram showing radiative (arrows), non-radiative transitions (zig-zag arrows), and energy transfer upconversion (ETU), phonon-assisted energy transfer (PAET), and excited-state absorption (ESA) channels for Stokes and upconverted emissions in $\text{Nd}^{3+}/\text{Yb}^{3+}$ doped nano-garnets.

Yb^{3+} absorption peaks are slightly shifted towards lower energy region upon reducing the size of RE^{3+} ($=\text{Gd}^{3+}$, Y^{3+} and Lu^{3+}) ion.

3.3. Luminescence

The luminescence spectra of $\text{Nd}^{3+}\text{-Yb}^{3+}$ co-doped REGG nano-garnets upon pumping Nd^{3+} ions with 805 nm laser excitation in resonance with the $^4\text{I}_{9/2} \rightarrow (^2\text{H}_{9/2}, ^4\text{F}_{5/2})$ transition were measured and compared with Nd^{3+} single-doped nano-garnets (see Fig. 5). After a fast non-radiative de-excitation to the $^4\text{F}_{3/2}$ emitting level, the 1 mol% Nd^{3+} doped REGG nano-garnets exhibit three main NIR bands in the 860–960 nm range, associated with the $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{9/2}$ transition, in the 1050–1130 nm range, to the $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{11/2}$ transition, and in the 1310–1430 nm range, related to the $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{13/2}$ transition.⁶ On the other side, co-doped nano-garnets exhibit an intense NIR band around 1025 nm associated with the $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ transition of Yb^{3+} ions due to phonon-assisted energy transfer (PAET) from Nd^{3+} ions to Yb^{3+} ions through the $(^4\text{F}_{3/2}; ^2\text{F}_{7/2}) \rightarrow (^4\text{I}_{9/2,11/2}; ^2\text{F}_{5/2})$ channel (see Fig. 5). The quasi-resonant overlapping of Nd^{3+} : $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{9/2}$ emission and Yb^{3+} : $^2\text{F}_{7/2} \rightarrow ^2\text{F}_{5/2}$ absorption is shown in Fig. 6.

The crystal-field splitting between $^2\text{F}_{5/2}$ and $^2\text{F}_{7/2}$ levels of Yb^{3+} in $\text{Nd}^{3+}\text{-Yb}^{3+}$:REGG nano-garnets (1000 cm^{-1}) is larger than in Yb^{3+} :YAG crystal (932 cm^{-1}), which favours for 1025 nm emission through $^2\text{F}_{5/2}(5) \rightarrow ^2\text{F}_{7/2}(2)$ transition, while the $^2\text{F}_{5/2}(5) \rightarrow ^2\text{F}_{7/2}(1)$ transition is diminished by a very strong re-absorption due to the overlapping of the absorption peaks. The higher absorption cross-section of Nd^{3+} ions at 805 nm in garnets ($8.3 \times 10^{-20}\text{ cm}^2$) is an indication for higher pumping efficiency of Yb^{3+} ions.²⁰ This sensitization was observed in materials: YAG nanocrystalline ceramics, YAlO_3 and $\text{YAl}_3(\text{BO}_3)_4$ crystals.^{21,25,27} The disappearance of Nd^{3+} emission in 10 mol% Yb^{3+} co-doped nano-garnets and the intense emission from the latter ion represents efficient energy transfer from Nd^{3+} to Yb^{3+} ions. However, $\text{Yb}^{3+} \rightarrow \text{Nd}^{3+}$ energy back transfer may not be

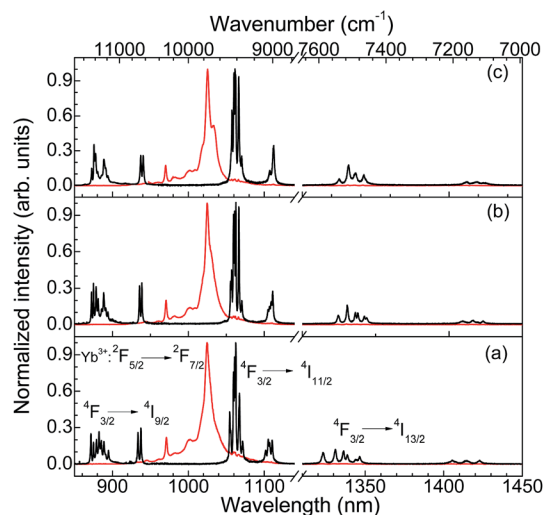


Fig. 5 Luminescence spectra of 1 Nd^{3+} singly doped (black) and 1 Nd^{3+} –10 Yb^{3+} co-doped (red) (a) GGG, (b) YGG and (c) LuGG nano-garnets ($\lambda_{\text{exc}} = 805\text{ nm}$).

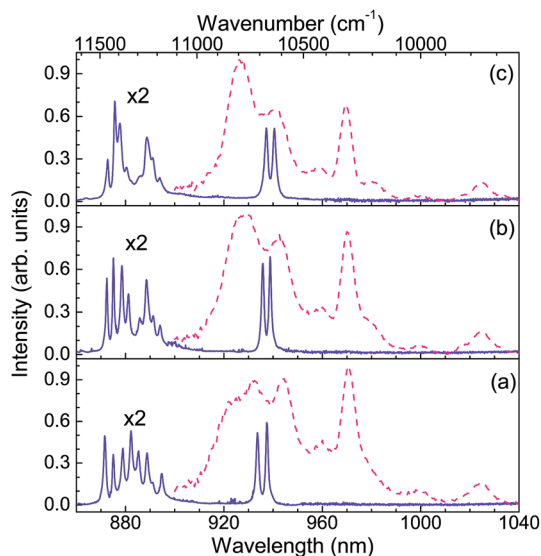


Fig. 6 Overlapping of Nd³⁺ emission (⁴F_{3/2} → ⁴I_{9/2}; solid line) and Yb³⁺ absorption (²F_{7/2} → ²F_{5/2}; dotted line) spectra of Nd³⁺/Yb³⁺ doped (a) GGG, (b) YGG and (c) LuGG nano-garnets.

possible because of weak overlapping between Yb³⁺ emission and Nd³⁺ absorption bands, which would need the absorption of a phonon.²¹

3.4. Decay curves

The luminescence decay curves of the ⁴F_{3/2} level in Nd³⁺–Yb³⁺ doped REGG nano-garnets were measured with a laser excitation at 532 nm in resonance with the ⁴I_{9/2} → ⁴G_{9/2} absorption and monitoring the ⁴F_{3/2} → ⁴I_{9/2} emission at 888 nm (see Fig. 7). Decay curves exhibit non-exponential nature and they have been analyzed considering the energy transfer between like centers. In this framework, the average distance *R* between Nd³⁺ and Yb³⁺ ions can be expressed as:³⁷

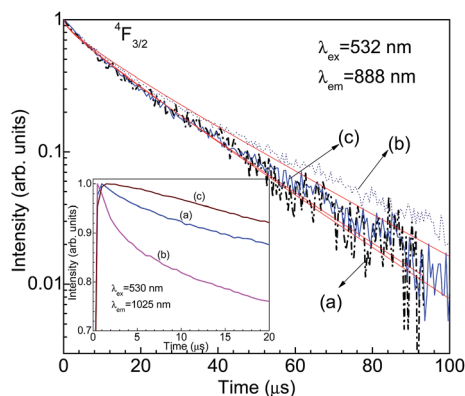


Fig. 7 Decay curves of the ⁴F_{3/2} level of Nd³⁺ ions in (a) YGG1Nd10Yb, (b) GGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets. The Inokuti–Hirayama fitting (solid line) for *S* = 6 is shown for all nano-garnets. Inset shows kinetic response of Yb³⁺ luminescence for all nano-garnets (λ_{ex} = 530 nm, λ_{em} = 1025 nm).

$$R = 2 \left(\frac{3V}{4\pi C_{x+y}N} \right)^{1/3} \quad (1)$$

where *N* is the number of sites that RE³⁺ ion can occupy per unit cell (for garnets *N* = 24), *V* is the volume of the unit cell, and *C_{x+y}* is the total doping concentration of *x* (Nd³⁺) and *y* (Yb³⁺) ions.

The distance *R* is found to be around 11.6, 11.2 and 11.2 Å for YGG1Nd10Yb, GGG1Nd10Yb and LuGG1Nd10Yb nano-garnets, respectively, which are similar to the reported value of around 11.6 Å between Nd³⁺ and Yb³⁺ ions in YAl₃(BO₃)₄ phosphors.³⁸ The energy transfer from a sensitizer to an activator may take place *via* exchange interaction and multipolar interaction. The mechanism of exchange interaction plays no role in energy transfer between Nd³⁺ or Nd³⁺–Yb³⁺ ions in these nano-garnets since it is dominant only for short distances (<6 Å). Hence, in the present case, the energy transfer between dopants may be due to electric multipolar interaction only.³⁹

The electric multipolar interaction can be understood by fitting the decay curves to the Inokuti–Hirayama (IH) model.⁴⁰ According to this model, the luminescence decay intensity *I*(*t*) after laser pulse excitation is given by

$$I(t) = I_0 \exp \left[- \left(\frac{t}{\tau_0} \right) - Q \left(\frac{t}{\tau_0} \right)^{3/S} \right] \quad (2)$$

where *τ*₀ is the intrinsic lifetime of the emitting level; *S* = 6, 8 or 10 depending on whether the dominant mechanism of the interaction between Nd³⁺ and Yb³⁺ ions is dipole–dipole, dipole–quadrupole or quadrupole–quadrupole, respectively; and the energy transfer parameter *Q* is defined as:

$$Q = \frac{4\pi}{3} \Gamma \left(1 - \frac{3}{S} \right) N_0 R_0^3 \quad (3)$$

where *N*₀ is the concentration of dopant; and *R*₀ is the critical transfer distance.

Decay curves have been fitted to IH model by varying *τ*₀ and *Q*. It is found that the decay curves are well fitted to *S* = 6, indicating that the dominant interaction between Nd³⁺ and Yb³⁺ ions are of dipole–dipole type. The resultant fitting parameters are shown in Table 2. Similar results were reported for 1 mol% Nd³⁺ single-doped YGG, GGG and LuGG nano-garnets with *τ*₀ (⁴F_{3/2}) as 315, 290 and 275 μs, respectively.⁶ Upon 10 mol% Yb³⁺ doping into Nd³⁺:REGG nano-garnets,

Table 2 Intrinsic lifetimes and energy transfer parameter for ⁴F_{3/2} and ⁴G_{7/2} levels of Nd³⁺ ions and ²F_{5/2} level of Yb³⁺ ion in 1Nd³⁺–10Yb³⁺ doped REGG nano-garnets

System	⁴ F _{3/2} (Nd ³⁺)		⁴ G _{7/2} (Nd ³⁺)		² F _{5/2} (Yb ³⁺)		Ref.
	<i>τ</i> ₀ (μs)	<i>Q</i>	<i>τ</i> ₀ (μs)	<i>Q</i>	<i>τ</i> ₀ (μs)	<i>Q</i>	
YGG1Nd10Yb	25	0.47	430	1.64	1110	0.95	Present
GGG1Nd10Yb	30	0.51	365	1.50	1000	0.94	Present
LuGG1Nd10Yb	24	0.39	330	1.06	625	0.55	Present
YGG1Nd	315	0.29	—	—	—	—	6
GGG1Nd	290	0.34	—	—	—	—	6
LuGG1Nd	275	0.36	—	—	—	—	6

drastic decrease of τ_0 from 315 to 25 μs , 290 to 30 μs , 275 to 24 μs is observed in YGG, GGG and LuGG nano-garnets, respectively, due to efficient $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer, shown in Table 2.

From the luminescence decay curves, the $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer efficiency η_{ETE} can be calculated by using the equation

$$\eta_{\text{ETE}} = 1 - \frac{\tau_{\text{SA}}}{\tau_{\text{S}}} \quad (4)$$

where τ_{SA} is the lifetime of sensitizer (S , Nd^{3+}) in the presence of activator (A , Yb^{3+}) and τ_{S} is the lifetime of sensitizer in absence of activator. The η_{ETE} is found to be around 90% for the three YGG1Nd10Yb, GGG1Nd10Yb and LuGG1Nd10Yb nano-garnets. Similar results were reported for Nd^{3+} – Yb^{3+} doped GGG crystal, YAG ceramics, Y_2O_3 ceramics, YAlO_3 , $\text{YAl}_3(\text{BO}_3)_4$ and $\text{Sr}_{0.6}\text{Ba}_{0.4}\text{Nb}_2\text{O}_6$ crystals and $\text{Al}_4\text{B}_2\text{O}_9$ nanocrystals (see Table 3 and Fig. 8).^{20–22,25,27,41} The higher transfer efficiencies are due to higher density of host cationic sites and the quasi-resonant superposition of the Nd^{3+} ($^4\text{F}_{3/2} \rightarrow ^4\text{I}_{9/2}$) emission and Yb^{3+} ($^2\text{F}_{7/2} \rightarrow ^2\text{F}_{5/2}$) absorption spectra (see Fig. 6).^{21,22} As shown in Fig. 8, the low value of η_{ETE} for $\text{Sr}_{0.6}\text{Ba}_{0.4}\text{Nb}_2\text{O}_6$ crystals may be due to unfavorable crystallographic structure of the host by substitution of divalent cation with trivalent cation.⁴¹ To confirm the energy transfer probability ($W_{\text{ET}} = 1/\tau_0$) of $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ channel, Stokes kinetic measurements were performed. Inset of Fig. 7 represents the normalized kinetic response for the Yb^{3+} luminescence detected at 1025 nm with a laser excitation at 530 nm. The fast rise time after a short pulse and a slow de-excitation are the fingerprints of energy transfer processes.⁴²

The luminescence decay curves of the $^2\text{F}_{5/2}(\text{Yb}^{3+})$ level in Nd^{3+} – Yb^{3+} co-doped REGG nano-garnets were measured by exciting at 920 nm and monitoring at 1025 nm. These decay curves are well fitted to the IH model for $S = 6$ (see Fig. 9). The energy transfer parameters are reported in Table 2 along with intrinsic lifetimes of the $^2\text{F}_{5/2}(\text{Yb}^{3+})$ level. In Yb^{3+} singly doped YGG, GGG and LuGG garnets, τ_0 ($^2\text{F}_{5/2}$) was reported to be around 1780, 1560 and 1040 μs , respectively.⁴³ Maia *et al.*, reported that the Yb^{3+} ($^2\text{F}_{5/2}$) lifetime was quenched from 176 to 15 μs upon increase of the Yb^{3+} concentration from 1 to 8 mol% in $\text{Al}_4\text{B}_2\text{O}_9$ nanocrystals.²⁶ The drastic decrease of $^2\text{F}_{5/2}$ level

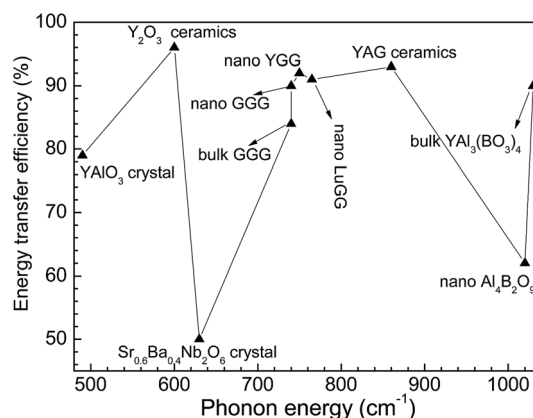


Fig. 8 Variation of $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer efficiency with phonon energy in different Nd^{3+} – Yb^{3+} co-doped matrices.

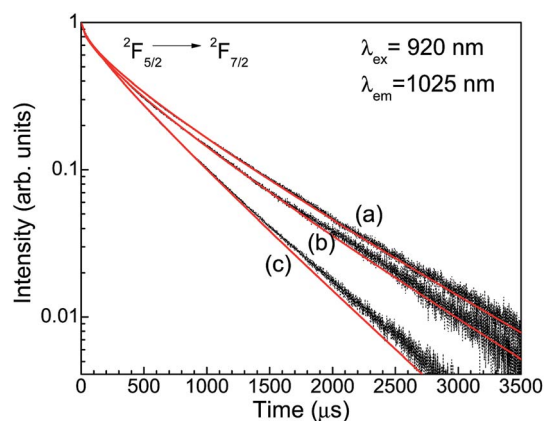


Fig. 9 Decay curves of $^2\text{F}_{5/2}$ level of Yb^{3+} ions in (a) YGG1Nd10Yb, (b) GGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets under resonant excitation. The IH fitting (solid line) for $S = 6$ is shown for all nano-garnets.

lifetime in Nd^{3+} – Yb^{3+} co-doped nano-garnets could be due to Yb^{3+} re-absorption with $[^2\text{F}_{5/2}(5), ^2\text{F}_{7/2}(1)] \rightarrow [^2\text{F}_{7/2}(1), ^2\text{F}_{5/2}(5)]$ channel. Similar results were reported for Nd^{3+} – Yb^{3+} doped GGG and YAG single crystals.^{20,21} It is worth noting that the longer luminescence lifetime for the $^2\text{F}_{5/2}$ level of Yb^{3+} ions in the present nano-garnets can reduce the pump threshold to get efficient emission at 1025 nm.

3.5. Upconversion

Upconversion (UC) is an optical process that converts two or more low energy photons into a high-energy photon, and this can be obtained by either excited state absorption (ESA) and/or energy transfer upconversion (ETU) processes. Under 970 nm laser excitation, intense green (520–560 nm), weak red (630–680 nm) and NIR (730–820 nm) upconverted emissions are observed corresponding to the $^4\text{G}_{7/2} \rightarrow ^4\text{I}_{9/2}$, $^4\text{G}_{7/2} \rightarrow ^4\text{I}_{13/2}$ and $^2\text{H}_{9/2} \rightarrow ^4\text{I}_{9/2}$ transitions, respectively (see Fig. 10). To analyze the UC processes, the upconverted emission intensities were recorded as a function of pump power. The upconverted emission

Table 3 Comparison of $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer efficiency (η_{ETE}) and phonon energies in different Nd^{3+} – Yb^{3+} co-doped matrices

Nd^{3+} – Yb^{3+} system	η_{ETE} (%)	Phonon energy (cm^{-1})	Ref.
YGG nanocrystal	92	750	Present
GGG nanocrystal	90	740	Present
LuGG nanocrystal	91	765	Present
GGG crystal	84	740	20
YAG ceramics	93	860	21
Y_2O_3 ceramics	96	600	22
YAlO_3 crystal	79	490	25
$\text{Al}_4\text{B}_2\text{O}_9$ nanocrystal	62	1020	26
$\text{YAl}_3(\text{BO}_3)_4$ crystal	90	1030	27
$\text{Sr}_{0.6}\text{Ba}_{0.4}\text{Nb}_2\text{O}_6$ crystal	50	630	41

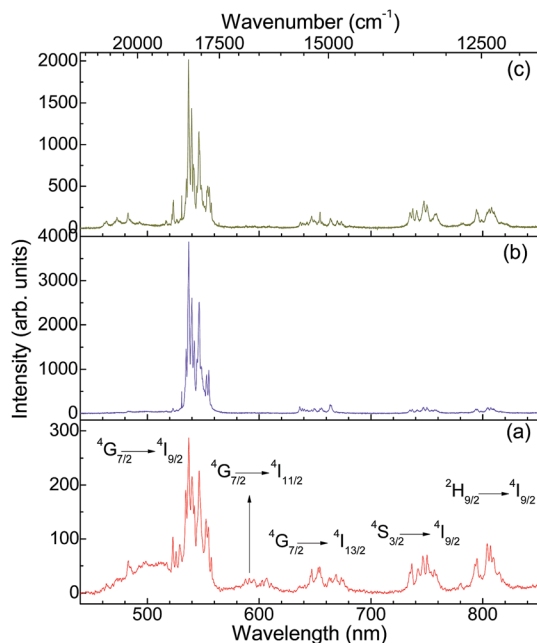


Fig. 10 Upconversion spectra of (a) GGG1Nd10Yb, (b) YGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets at $\lambda_{\text{exc}} = 970$ nm with laser power of 610 mW.

intensity (I_{UP}) follows the relation: $I_{\text{UP}} \propto (I_{\text{IR}})^n$, where ' I_{IR} ' is the IR excitation intensity, ' n ' is the number of IR photons absorbed per emitted photon. As shown in Fig. 11, the slope values are found to deviate from expected value of 2, which could be due to infinitely large UC rates.⁴⁴

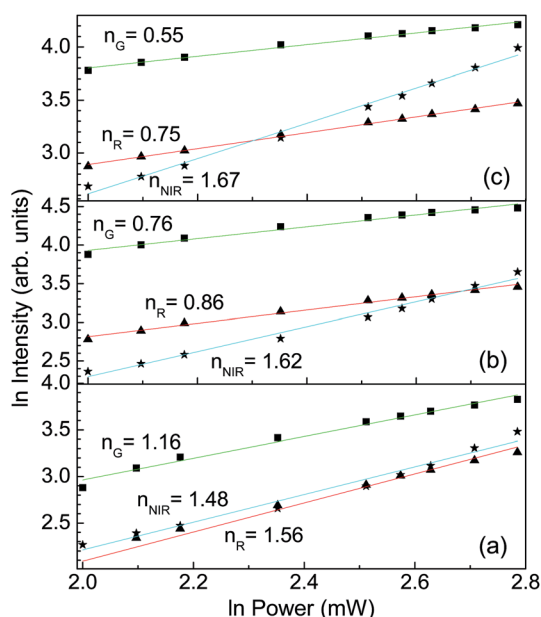


Fig. 11 Pump power dependence of the green, red and NIR upconverted emission intensities of Nd^{3+} ions in (a) GGG1Nd10Yb, (b) YGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets (n_{G} , n_{R} and n_{NIR} represents slope for green, red and NIR emissions, respectively).

The upconverted emission from Nd^{3+} ions and a simple schematic representation of Yb^{3+} to Nd^{3+} energy transfer processes are depicted in Fig. 4. Following the absorption of 970 nm pump photon, the Yb^{3+} ions are excited from the $^2\text{F}_{7/2}$ ground state to the $^2\text{F}_{5/2}$ excited level, then non-resonantly transfers its energy (ETU1) to the Nd^{3+} ions and excites them from the $^4\text{I}_{9/2}$ level to the $^4\text{F}_{3/2}$ one through phonon-assisted energy transfer. The Nd^{3+} ions are consequently and resonantly excited to $^2\text{G}_{9/2}$ state *via* ETU2, and then decay non-radiatively to the $^4\text{G}_{7/2}$ level. This in turn leads to the intense green ($^4\text{G}_{7/2} \rightarrow ^4\text{I}_{9/2}$), weak red ($^4\text{G}_{7/2} \rightarrow ^4\text{I}_{13/2}$) and NIR ($^2\text{H}_{9/2} \rightarrow ^4\text{I}_{9/2}$) transitions.

There is another possible channel to populate $^4\text{G}_{7/2}$ level of Nd^{3+} ions *via* ESA mechanism. A 970 nm laser photon may excite the Nd^{3+} ions from $^4\text{I}_{9/2}$ to $^4\text{F}_{3/2}$ level through phonon assisted energy transfer. Consequently, within the same laser pulse, Nd^{3+} ion absorbs a second photon and reaches the $^2\text{G}_{9/2}$ level. As shown in Fig. 4, the $^4\text{G}_{7/2}$, ($^4\text{F}_{7/2}$, $^4\text{S}_{3/2}$) and ($^4\text{F}_{5/2}$, $^2\text{H}_{9/2}$) levels are populated through multiphonon relaxation process and emit intense green, weak red and NIR emissions from the respective emitting levels. This can be understood by analyzing the ETU/ESA contributions, the involved metastable levels, and the dynamics of the green emitting level.

Luminescence decay curves for green emission ($^4\text{G}_{7/2}$ level) of Nd^{3+} ion in all three co-doped nano-garnets have also been measured under resonant excitation to evaluate ESA/ETU contributions by comparing temporal evolution of green emission under 970 nm excitation. Moreover, the τ_0 of $^4\text{G}_{7/2}$ (Nd^{3+}) level is found to be around 430, 365 and 330 μs in $1\text{Nd}^{3+}\text{-}10\text{Yb}^{3+}$ -doped YGG, GGG and LuGG nano-garnets, respectively, measured under resonant excitation at 488 nm by monitoring the emission at 537 nm. The decay curves are also well-fitted to IH model for $S = 6$ (not shown in figure) and the intrinsic lifetime of $^4\text{G}_{7/2}$ level and energy transfer parameters are determined and are given in Table 2. There were only few reports found on evaluation of $^4\text{G}_{7/2}$ level lifetime that include Nd^{3+} singly doped KPb_2Cl_5 and KPb_2Br_5 crystals having 55 and 63 μs , respectively.^{45,46}

3.6. Dynamics

In order to estimate the ESA/ETU contributions involved in obtaining UC green emission, the temporal evolution of the upconverted green emission is measured at 970 nm and compared with decay curves measured under 488 nm excitation, resonant to the $^4\text{I}_{9/2} \rightarrow ^4\text{G}_{7/2}$ transition (see Fig. 12). If the temporal evolution of the green UC luminescence is exactly the same as the luminescence decay curve obtained at resonant excitation, the observed green UC emission should be due to ESA mechanism.⁴⁷ As can be seen from Fig. 12, the observed green UC luminescence is not only due to ESA but also due to other mechanisms. Apart from the ESA mechanism, energy transfer processes between nearby Nd^{3+} or Yb^{3+} ions in the ground ($^4\text{I}_{9/2}$) and intermediate ($^4\text{F}_{3/2}$ and $^2\text{G}_{9/2}$) states could take place if the ETU mechanism dominates the UC process. To calculate quantitative values of ESA and ETU contributions in green UC emission, the temporal evolution of the upconverted

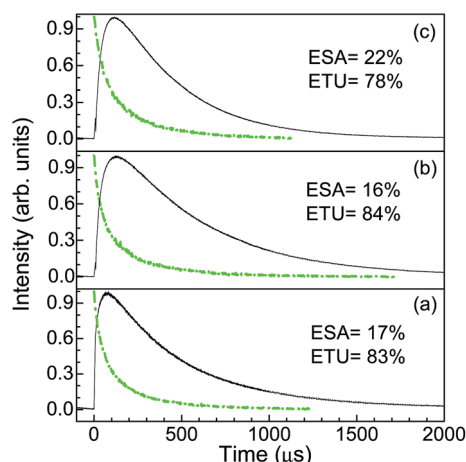


Fig. 12 Temporal evolution of the green up-converted luminescence of Nd^{3+} ions in (a) GGG1Nd10Yb, (b) YGG1Nd10Yb and (c) LuGG1Nd10Yb nano-garnets under 970 nm excitation. The luminescence decay curves of the green emission (dotted) under direct excitation to the $^4\text{G}_{7/2}$ level at 488 nm are also included for comparison.

green intensity is superimposed on the resonantly excited Stokes luminescence decay curve and calculated the area under these curves. The ESA contribution is evaluated by estimating the part of decay area over total area of temporal evolution, and the rest of value is ETU's part.⁴⁷ The ESA & ETU's contributions are found as 17 & 83% for GGG1Nd10Yb, 16 & 84% for YGG1Nd10Yb and 22 & 78% for LuGG1Nd10Yb nano-garnets, respectively, indicating that the mechanisms are quite similar for the studied nano-garnets.

4. Conclusions

The $\text{RE}_3\text{Ga}_5\text{O}_{12}$ (RE = Gd, Y and Lu) nano-garnets co-doped with 1Nd^{3+} – 10Yb^{3+} ions are synthesized in single phase of garnet structure with average crystallite size of about 35 nm. Upon exciting the Nd^{3+} ions at 805 nm, an intense near infrared emission from the Yb^{3+} ions is observed at around 1025 nm. The lifetime of $^4\text{F}_{3/2}$ (Nd^{3+}) level decreases drastically upon co-doping with 10 mol% of Yb^{3+} ions with 1 mol% of Nd^{3+} ions, which is due to efficient $\text{Nd}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfer process. Moreover, lifetimes of the $^4\text{G}_{7/2}$ (Nd^{3+}) and the $^2\text{F}_{5/2}$ (Yb^{3+}) levels are evaluated. These decay curves are well-fitted to the Inokuti–Hirayama model for $S = 6$, indicating that interaction between Nd^{3+} and/or Yb^{3+} ions is of dipole–dipole. The upconversion luminescence of 1Nd^{3+} – 10Yb^{3+} co-doped $\text{RE}_3\text{Ga}_5\text{O}_{12}$ nano-garnets exhibits intense green emission due to both phonon assisted excited state absorption and energy transfer upconversion mechanisms. These results indicate that these gallium-based nano-garnets could find applications in ultra-fast lasers, solar energy converters and phosphors.

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